Vitor Pinheiro

AI/ML Software Engineer
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SUMMARY

AI/ML engineer with 5 years shipping ML, LLM and data systems in fintech (lending, credit risk, fraud) and energy forecasting. Recent work includes a LightGBM fraud model that cut false-positive alerts by 99% for a US bank, an LLM pipeline that explains each fraud alert to analysts in plain language, and a LangGraph support agent that resolves 80% of internal tickets. Strong in Python, Spark and Databricks end to end, from data pipelines and MLOps to model serving and agents. Based in Brazil in UTC-4, the same time zone as US Eastern for most of the year, and open to remote work.

SKILLS

- **Languages:** Python, Scala, SQL, Clojure, JavaScript/TypeScript, HTML, CSS
- **Machine learning:** LightGBM, XGBoost, SHAP, Keras, TensorFlow, OR-Tools, time-series forecasting, experimentation
- **LLMs and agents:** LangChain, LangGraph, OpenAI, Mosaic AI Model Serving, LLM evaluation, Claude, Cursor
- **Data and MLOps:** Spark/PySpark, Databricks, Delta Live Tables, Unity Catalog, MLflow, Airflow, Streamlit, FastAPI
- **Cloud and DevOps:** Azure, Docker, Kubernetes, GitHub Actions (CI/CD), observability
- **Domains:** credit and lending (NPV, offer experimentation, policy datasets), fraud detection and transaction monitoring, time-series forecasting, LLM agents, MLOps

EXPERIENCE

Qubika · Senior AI/ML Engineer (client: Coastal Community Bank, US)

Remote, part-time · Feb 2026 – Jul 2026

- Productionized a LightGBM fraud detection model that, on a 24-month backtest, cut transaction-monitoring alert volume by 99.2% (12.9M to 98.5K) and lifted analyst-facing precision from 1.29% to 76.4%, a 131x reduction in review burden while still surfacing 45% of confirmed fraud.
- Took analyst load from about 538K alerts a month at 78 alerts per confirmed fraud to about 4.1K alerts a month at 1.3 alerts per confirmed fraud, a 60x efficiency gain.
- Designed and shipped the end-to-end Databricks platform behind the model: a Delta Live Tables pipeline materializing segment-specific feature tables in Unity Catalog, and a Databricks Asset Bundle orchestrating training, scoring and promotion as version-controlled jobs, with MLflow tracking and models served from the Unity Catalog Model Registry across environments.
- Built an explainability pipeline that turns each fraud alert into a reviewer-ready rationale by computing SHAP attributions and calling a Databricks-hosted foundation model through Mosaic AI Model Serving with structured prompts, producing customer- and transaction-context narratives that speed up case triage.

Nubank · AI/ML Software Engineer

São Paulo, BR (hybrid) · Sep 2022 – Jul 2026 · promoted from Intern to Engineer

AI and LLM systems

- Built a Slack AI assistant with Python, LangChain, LangGraph and OpenAI over the framework's codebase, documentation and Slack channel history, running as a FastAPI service on Docker and Kubernetes. It resolves 80% of support tickets without a human (10 to 2 a week), freeing engineering time for infrastructure and features.
- Prompted, tuned and evaluated Databricks Genie as a self-service analytics assistant for the Lending business unit, defining business concepts and field-level metadata over core customer and loan datasets. The proof of concept with business users handled 3 of 5 test scenarios and surfaced the ETL and documentation gaps that shaped the adoption strategy.

Machine learning and experimentation

- Translated the Lending NPV framework from Scala to Python/PySpark, unifying profitability analysis across real-time credit models (loan eligibility and real-time valuation) with distributed and single-node simulations of loan performance across 50+ product lines.
- Co-designed Scala/Spark experimentation logic that randomized personal-loan offer parameters (installments, maximum principal, interest rates) on customer subsamples, reducing survivorship bias and producing 20%+ cleaner training data for credit policy decisions.
- Engineered critical features for credit risk models, improving performance through systematic feature-importance analysis and domain-driven transformations.

Data platform

- Reduced landing time for critical Lending core policy datasets by 34 hours and improved model feature freshness by 2 days by reordering ETL execution, pushing filters earlier in the Scala/Spark DAG and applying time-based partitioning.
- Co-developed a Scala data transformation framework with segment, metric and aggregation catalogs and a shared API, centralizing Lending business definitions, and automated .jar deployment to Databricks clusters with GitHub Actions.
- Led data governance initiatives for Lending, including the subdomain migration, SCR monitoring infrastructure and credit metrics catalog consolidation, moving 100% of the team onto the new infrastructure.
- Implemented data validation and monitoring pipelines that kept lending data reliable across systems.
- Implemented observability for the team's ML services and optimized CI/CD pipelines with GitHub Actions.

- Provided 24/7 support for the monitoring infrastructure, onboarding multiple business units with zero critical incidents.

Analytics tools

- Turned a ~2-day manual reporting process into a weekly task of under an hour by co-developing an internal low-code dashboard framework in Python/Streamlit, with reusable APIs for plots, filters and aggregations, S3 data loading, SQLite caching, CLI setup and python-pptx exports. Adoption grew from Lending to 19+ business units, and several teams replaced Looker at about 1 USD a day in infrastructure costs.
- Built analytics abstractions (filters, aggregations, re-aggregated monitorings) for the framework, cutting execution time from ~20 minutes to ~5 seconds, and a weekly KPI monitoring routine that standardized metrics across credit programs, which 12 business units used to automate their monitoring.
- Designed APIs over datasets and pre-aggregated monitoring data that let non-technical users build dashboards without code in a no-code dashboard builder MVP.
- As an intern, supported credit-risk and loan-performance model development, ran EDA on lending datasets, and wrote documentation and playbooks for ML workflows.

EloGroup · Data Science and Engineering Intern (client: Comgás)
São Paulo, BR (remote) · Jun 2021 – Sep 2022

- Helped cut natural-gas contractual fines by roughly R$200M in the first three months for Comgás, Brazil's largest natural-gas distributor. Co-implemented a LightGBM/SARIMAX/Holt-Winters ensemble over 24 city-level demand series, validated with ADF, KPSS, Granger and Ljung-Box tests, that kept 18 of 24 citygates within the 5% fine threshold in demand programming with Petrobras.
- Co-designed end-to-end ML pipelines in Azure ML with MLflow tracking, migrating a Databricks-only solution to a managed setup orchestrated by Azure Data Factory, from Azure Data Lake to Synapse views and Power BI dashboards, with automated data quality checks in a medallion architecture.
- Built a Streamlit app that automated a Box-Jenkins-style forecasting workflow, from visualization to model evaluation, for business stakeholders.
- Standardized text processing across consulting projects by co-implementing an internal NLP framework in Streamlit with spaCy and NLTK.

PROJECTS

- **CarDeck** (github.com/jvitorcosta/spotify-car-deck), 2026: open-source ESP32 firmware that turns a dashboard screen into a Spotify now-playing display styled as a Pokémon battle, with synced lyrics, a companion screen and a Bluetooth OBD gauge cluster kept read-only by a tested allow-list. Has CI and tagged releases.
- **SolvedOR** (github.com/jvitorcosta/or-solver-streamlit), bachelor thesis: a Streamlit app that teaches Operations Research. Students write models in a custom syntax, OR-Tools solves them, and Plotly shows the feasible region in up to 3D.

EDUCATION

Bachelor of Industrial Engineering, Universidade Federal do Amazonas (UFAM), Manaus, BR · Jan 2019 – Sep 2023 · GPA 9.07/10

- Relevant coursework: Operations Research, Simulation, Numerical Analysis, Probability and Statistics, Time Series, Optimization, Stochastic Processes, Data Analysis.

CERTIFICATES

- Google Data Analytics Professional Certificate, Coursera · June 2023
- Information Technology Technician, Matias Machline Foundation · December 2016

LANGUAGES

Portuguese (native) · English (advanced, C1) · Mandarin (intermediate, B1)